

# Model Based Photopic Electroretinogram Source Separation: A Multiresolution Analysis Approach

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**Abstract**— Visual function testing using electroretinogram (ERG) signal is used to detect abnormalities of the retina. This is accomplished by measuring, characterizing and analyzing the bio potential responses from various retinal cells formed due to a visual stimulation. The objective of this study is to improve the existing ERG signal characterization models by identifying and including a key component called i-wave in the photopic response. To validate the proposed characterization model, a signal analysis and processing algorithm based on Multi Resolution Analysis has been developed to reliably separate various underlying components of this signal. Finally, the fidelity of this separation is quantitatively and qualitatively assessed by computing the Pearson's correlation coefficient and the corresponding scatter plots between the composite and the reconstructed ERG signal.

**Keywords:** Medical engineering, electroretinogram, ERG, i-wave, photopic response, multiresolution analysis, wavelet source separation.

## I. INTRODUCTION

Analyzing the electroretinogram (ERG) signal to predict the presence of retinal dysfunctions and diseases is a promising visual function evaluation procedure [1, 2]. Visual function evaluation can also be accomplished by technologies like electrooculogram (EOG) and visually evoked potentials (VEP). However, ERG signal contains more information on the pathologies of the visual system in comparison to EOG and VEP. This provides the motivation for characterizing and analyzing the ERG signal.

The full-field ERG signal is a mass recording of bio potential responses from various retinal cells formed due to a visual stimulation from a bright flash light. Based on the conditions under which the visual stimulation is initiated, a corresponding response is generated by the retina. Literature [3, 4, 5] shows that the vast majority of ERG responses include a-wave, and b-wave. In addition, an ERG response generated under selective visual stimulation conditions can also reveal the presence of other components such as Oscillatory Potentials (OPs), Photopic Negative Response (PhNR) and i-wave that can provide significant diagnostic information on retinal and optical nerve diseases.

The a-wave represents the response of photoreceptors to light [6]. The b-wave, on the other hand, mostly represents the activity of depolarizing (ON) bipolar cells [7]. Furthermore, the OPs, which originate in the inner retina can provide useful information about the progression and treatment efficacy in

diabetic retinopathy and other retinal disorders [8, 9]. It has been shown by Frishman *et al.* [10] that PhNR generated by retinal ganglion cell activity can potentially reveal the presence of glaucoma. The photopic ERG signal also consists of another key component called i-wave, observed by Rosolen *et al.* [11]. The i-wave is likely generated at the optic nerve, which, if proved accurate, would have a significant diagnostic implication for a number of retinal and optic nerve diseases. Due to uncertainty in the precise origin of i-wave in humans, the existing ERG characterization models did not recognize the presence of this key component in the photopic response.

Granit [12] characterized an ERG response as a summation of a-wave, b-wave, and PhNR and separated them pharmacologically. Alfieri *et al.* [3] characterized a photopic ERG signal as a summation of a-wave, b-wave, and OPs and used Fourier series to separate the OPs from the composite signal. Castro *et al.* [4] empirically modelled a photopic ERG signal as a summation of a-wave and b-wave based on the light intensity and latency. No other comparable studies have accomplished characterization, analysis and source separation of a photopic ERG signal. In practice, a full-field ERG signal is short lived mainly because of two reasons. First, the duration of the flash light stimulus is typically short enough ( $\leq 4$  ms) to limit the retinal response to a smaller duration ( $\leq 0.1$  s). Second, due to the occurrence of photomyoclonic reflex that interferes with the recovery of the ERG signal to baseline [13]. Thus, because of its relatively short duration, conditional presence of the components and low signal to noise ratio, characterization and analysis of an ERG signal is still a challenge.

The objective of this study is to not only identify and incorporate i-wave into the photopic ERG response but also to reliably separate various underlying components and provide significant diagnostic information regarding retinal and optical nerve diseases from them. To accomplish this objective, an improved ERG signal characterization model was proposed with the assumption that an ERG signal is a summation of bio potential responses from discrete depths and cell populations of the retina. Then the proposed characterization model was validated by separating various components of the photopic human ERG signal using an approach based on Multi Resolution Analysis (MRA). Finally, a quantitative and a qualitative measure of fidelity of this separation is obtained by means of computing the Pearson's correlation coefficient and by obtaining scatter plots between the composite photopic ERG and the reconstructed ERG.

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Section 2 describes how data was collected for the current study. Section 3 explains the signal analysis and processing algorithm that was developed to separate all the underlying components of the ERG signal. Section 4 presents the results obtained from the signal analysis and processing algorithm in detail. Section 5 not only highlights the existing challenges with respect signal analysis and processing but also proposes potential directions for future work. Section 6 provides a summary of all aspects involved in this study.

## II. DATA COLLECTION

To collect the ERG data, the subject's eye is visually stimulated with a brief flash light ( $\leq 4$  ms) under a relatively bright background ( $\geq 30$  cd/m<sup>2</sup>) for a specific onset time and a corresponding bio potential response is acquired by means of surface electrodes connected to the subject's cornea. Depending on the intensity and wavelength of the stimulus light, the background light and its corresponding onset time pattern, either a scotopic or a photopic ERG response is acquired. Scotopic ERG is a response predominantly originating from the rod photoreceptors of the retina; photopic ERG is a response predominantly originating from the cone photoreceptors of the retina. The individual components of the ERG are directly associated with the visual stimulation pattern [14].

A photopic ERG signal was obtained in this study from both the left and the right eyes of a healthy male subject in his mid-fifties. Signal was acquired from cornea using surface thread electrodes, where DTL Plus Electrode™ [15] was the active electrode and Ag/AgCl skin electrodes were placed below the lower eyelid and on the skin overlying the left mastoid as reference and as ground electrodes respectively. The retina was stimulated by means of a brief red flash light ( $<1$  ms,  $\sim 2.5$  cd/m<sup>2</sup>) generated under a white background (30 cd/m<sup>2</sup>). The resulting ERG signal was band pass filtered over 0.3 – 500 Hz, ensemble averaged over ten trials, digitized (2000 Hz sampling rate and 16-bit quantization) and stored using UTAS E-3000 system [16]. Figures 1a and 1b show the photopic ERG signal obtained from both left and right eyes.

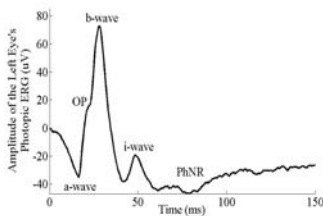


Figure 1a: Photopic ERG Signal From Left Eye.

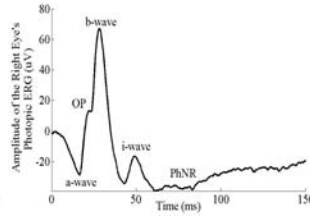


Figure 1b: Photopic ERG Signal From Right Eye.

## III. SIGNAL ANALYSIS AND PROCESSING

Characterization and analysis of the photopic ERG response was seen as a problem that requires a reliable source separation. In order to accomplish the separation of the components that make up photopic ERG signal, an improved characterization model that incorporates a new component called i-wave into the photopic response was assumed as an underlying photopic ERG signal characterization model. Equation 1 presents the proposed characterization model.

$$\text{ERG} = P_1 + P_2 + P_3 + \text{OPs} + \text{PhNR} \quad (1)$$

Where  $P_1$  represents the a-wave,  $P_2$  represents the b-wave,  $P_3$  represents the i-wave, OPs represents the oscillatory potentials and PhNR represents the photopic negative response. The separation process is initiated by removing a template of a-wave and b-wave developed using the mathematical model presented in [4] from the composite photopic ERG signal. Then, using a wavelet based Multiresolution Analysis [17], OPs, PhNR and i-wave are separated from the residual signal. Figure 2 shows the signal analysis and processing algorithm that was developed to accomplish the source separation and validation.

In MRA, the given signal  $\hat{x}(n)$ , i.e., the residual signal obtained after removal of the a-wave and the b-wave is decomposed into various levels of Approximation (A) and Detail (D) coefficients according to (2).

$$\begin{aligned} A_m(n) &= \langle \hat{x}(n), \varphi_{mk}(n) \rangle \\ D_m(n) &= \langle \hat{x}(n), \psi_{mk}(n) \rangle \end{aligned} \quad (2)$$

Where  $\langle \rangle$  operator represents the inner product,  $m$  represents the decomposition level,  $k$  represents the translation,  $\psi$  represents the mother wavelet with  $R$  vanishing moments and  $\varphi$  corresponds to its scaling function [18].

The approximation and detail coefficients were obtained by hard threshold of MRA coefficients based on the underlying component and reconstructed back into time domain. The chosen mother wavelet ( $\psi$ ) should provide a reasonably good low and high frequency resolution to the underlying components of  $\hat{x}(n)$ , through compact support. Additionally, the residual signal has shown the presence of discontinuities in its higher order derivatives. As a result, the chosen  $\psi$  need to be able to detect the presence of hidden discontinuities. Finally,  $\psi$  should be orthogonal to avoid phase distortions from the transformation. All the requirements of the current study were satisfied by the Daubechies wavelet with ten vanishing moments (db10). Also, it has to be noted that the shape of Daubechies wavelet qualitatively has a very close resemblance to a-wave, b-wave, i-wave and OPs.

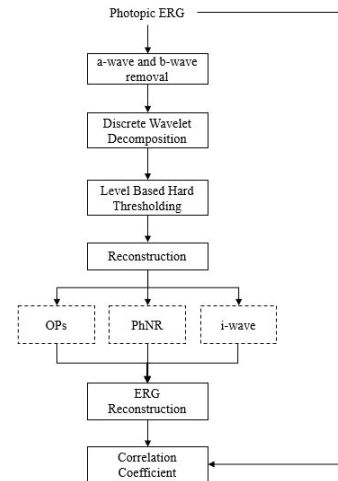


Figure 2: Photopic ERG Source Separation and Validation Algorithm.

#### IV. RESULTS

It was observed that a-wave, b-wave, and i-wave correspond to a similar frequency range as shown in Figure 3 which depicts a contour plot of continuous time wavelet transform (CTWT) representation for the photopic ERG signal of the right eye obtained using db10 wavelet. As a result, the multiresolution analysis was not able to reliably isolate the a-wave, b-wave and i-wave from the composite photopic ERG signal.

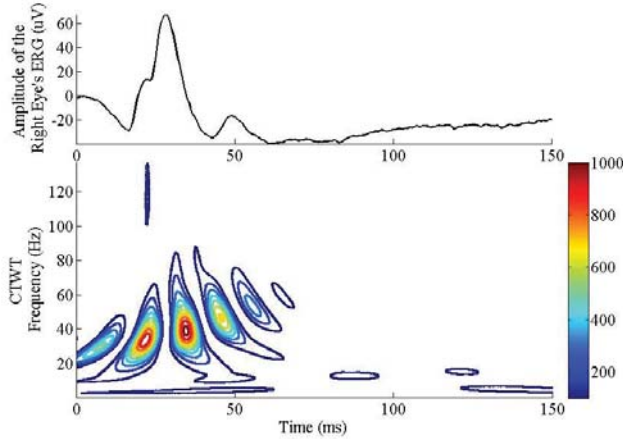


Figure 3: Continuous Time Wavelet Transform Representation of Right Eye's Photopic ERG.

Hence, a combined template of a-wave and b-wave was developed based on peak amplitude, shape of the waveform and removed from the composite ERG signal. The resulting residual signal is subjected to a six level MRA using db10 wavelet. The OPs were extracted by performing a hard threshold on MRA coefficients and retaining Detail coefficients at levels one (D1), two (D2) and three (D3) simultaneously and reconstructing them back into time domain. The PhNR was extracted by performing a hard threshold on MRA coefficients and retaining Approximation coefficients at level six (A6) and reconstructing them back into time domain. Finally, the i-wave was extracted by removing the OPs and PhNR from the residual signal. Figure 4 shows the MRA decomposition of the residual signal.

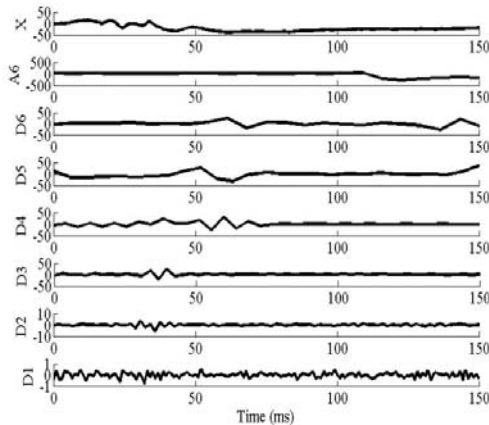


Figure 4: MRA Decomposition to Separate OPs and PhNR From the Residual Signal.

Figures 5a-5d show the residual signal, OPs, PhNR, and i-wave respectively separated from the ERG for the right eye. Finally, a reconstructed ERG signal was computed by superimposing all the separated components.

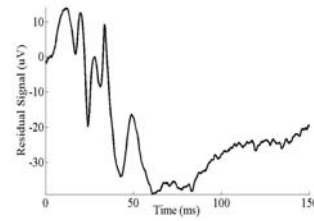


Figure 5a: Residual Signal

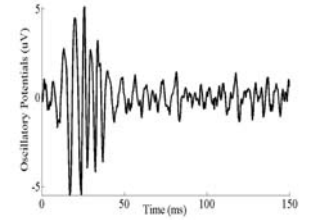


Figure 5b: Oscillatory Potentials

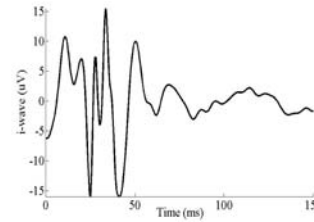


Figure 5c: i-wave

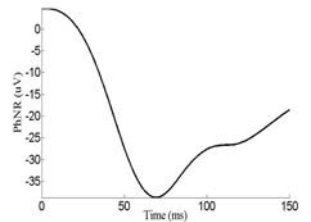


Figure 5d: Photopic Negative Response

Similar results have been obtained by applying the same signal analysis and processing to the left eye's composite ERG. However due to page length restriction, only the validation results are presented here for the left eye. In order to assess the fidelity of the performed separation, qualitative and quantitative measures were employed in the current study. Scatter plots were obtained between the composite photopic ERG and the corresponding reconstructed ERG for both the left and right eyes as a qualitative measure; Pearson's correlation coefficient was computed between the composite ERG and the reconstructed ERG as a quantitative measure. Figures 6a and 6b provide a scatter plot obtained between the composite ERG and the reconstructed ERG for both the left and right eyes.

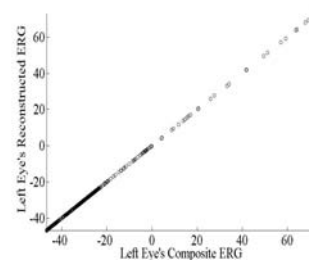


Figure 6a: Scatter Plot of Left Eye's Composite and Reconstructed ERG

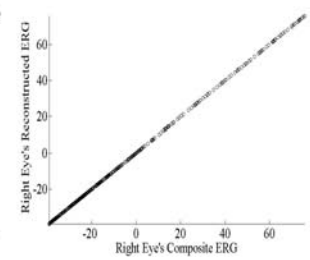


Figure 6b: Scatter Plot of Right Eye's Composite and Reconstructed ERG

The scatter plots exhibit complete qualitative correlation between the composite and the reconstructed ERG signal. Additionally, the computed Pearson's correlation coefficient also shows 100% percent correlation between the composite and the reconstructed ERG for both the left and right eyes and thus quantitatively assessing the performed source separation. The research work presented in this paper was in part documented as a Master's thesis at the University of South Florida [19].



## V. CHALLENGES AND FUTURE WORK

Results shown in this paper with respect to i-wave are considered to be preliminary at this point. Therefore, extending the current analysis to a statistically robust dataset and standardizing the identification of i-wave is left for future study. It was also found that not a single public database for clinically acquired ERG data is available. Thus, establishing an ERG database is another future work of this investigation. The current analysis has not considered the effect of channel noise on the data. In future analysis, a noise cancellation algorithm need to be employed to remove the channel noise from the data and thus providing a better source separation. Evaluating the results of techniques like principle component analysis and independent component analysis on the random electrical activity rather than the ensemble averaged data would be interesting. Performing a multimodal analysis of ERG signal along with an additional modality like visually evoked potentials should open up a new dimension in the ERG signal analysis and processing. Additionally, the accuracy of current source separation technique needs to be validated for multifocal ERG and various conditions of visual stimulation patterns with respect to the wavelength and intensity of the stimulus light, the background light and the onset time. In addition, evaluating the diagnostic relevance of the separated components with respect to pathologies like glaucoma and diabetic retinopathy is currently being investigated.

## VI. CONCLUSION

This study proposed an improved ERG signal characterization model by identifying and including a key component called i-wave into the photopic response. For the first time, i-wave and PhNR were characterized and analyzed as independent components of a composite photopic ERG response. Further, the proposed characterization model was validated by separating all the underlying components of the acquired response using MRA. Finally, the performed separation was qualitatively and quantitatively assessed by means of both the scatter plots and the Pearson's correlation coefficient.

## REFERENCES

- [1] A. Chow, M. Pardue, V. Chow, G. Peyman, C. Liang, J. Perlman and N. Peachey, "Implantation of Silicon Chip Microphotodiode Arrays into the Cat Subretinal Space," *IEEE Transactions On Neural Systems And Rehabilitation Engineering*, vol. 9, no. 1, pp. 86 - 95, 2001.
- [2] J. Weiland, G. Fuji, M. Mahadevappa, R. Greenber, M. Tameesh, D. Guven, E. de Juan Jr. and M. Humayun, "Chronic Electrical Stimulation of the Canine Retina," in *IEEE Engineering In Medicine and Biology*, Houston, USA, 2002.
- [3] R. Alfieri, E. Albuissou and J. Boire, "Fourier Series Applied To Oscillatory Potentials," in *IEEE Engineering In Medicine & Biology Society*, 1988.
- [4] J. Castro, A. Meyer and E. Haup, "Modeling Of Electroretinogram And Its Relation To Stimulus Light Intensity," in *IEEE Engineering in Medicine and Biology Society*, Orlando, USA, 1991.
- [5] C. Finelli, S. Doty, K. Svinarich and B. Winkler, "A Digitized Acquisition System for Electroretinograms and its Use in Studying the Effects of Organic Buffers on the Isolated Rat Retina," in *IEEE Engineering in Medicine and Biology Society*, Chicago, 1997.
- [6] J. Robson and L. Frishman, "The Rod-driven a-wave of the Dark-adapted Mammalian Electroretinogram," *Progress in Retinal and Eye Research*, vol. 39, pp. 1 - 22, 2014.
- [7] L. Frishman and M. Wang, "Electroretinogram of Human, Monkey and Mouse," in *Adler's Physiology of the Eye*, Saunders 2011, 2011, pp. 480 - 501.
- [8] R. Tzekov and G. Arden, "The Electroretinogram in Diabetic Retinopathy," *Survey of Ophthalmology*, vol. 44, no. 1, pp. 53 - 60, 1999.
- [9] L. Wachtmeister, "Oscillatory Potentials in the Retina: What Do They Reveal," *Progress in Retinal and Eye Research*, vol. 17, no. 4, pp. 485 - 521, 1998.
- [10] L. Frishman, S. Viswanathan and J. Robson, "The Photopic Negative Response of the Flash ERG: Changes in Experimental Glaucoma in Monkeys and Clinical Applications in Humans," in *IEEE Engineering in Medicine and Biology Society*, Houston, USA, 2002.
- [11] S. Rosolen, F. Rigaudière, J. LeGargasson, C. Chalié, M. Rufiange, J. Racine, S. Joly and P. Lachapelle, "Comparing the Photopic ERG i-wave in Different Species," *American College of Veterinary Ophthalmologists*, vol. 7, no. 3, pp. 189 - 192, 2004.
- [12] R. Granit, "The Components Of The Retinal Action Potential In Mammals And Their Relation To The Discharge In The Optic Nerve," *Journal Of Physiology*, vol. 77, no. 3, pp. 207 - 239, 1933.
- [13] M. Johnson and R. Massof, "The Photomyoclonic Reflex: An Artefact in the Clinical Electroretinogram," *British Journal of Ophthalmology*, vol. 66, pp. 368 - 378, 1982.
- [14] K. Davey, B. Thompson, S. Wang, A. Koblasz and B. Nation, "Predicting Distributed Retinal Source Activity from ERG Data-Part 11: Multiple Stimulus Approach," *IEEE Transactions On Biomedical Engineering*, vol. 35, no. 11, pp. 948 - 952, 1988.
- [15] Diagnosys LLC, "DTL Plus Electrode™," Diagnosys LLC, [Online]. Available: <http://diagnosysllc.com/products/dtl/>.
- [16] LKC Technologies, Inc., "Universal Testing And Analysis System - Electrophysiologic 3000 (UTAS-E 3000)," 2003. [Online]. Available: <http://www.lkc.com/downloads/User%20Manuals/UTAS-E%203000/UTAS-E%203000%20Service%20Manual%203.0.pdf>.
- [17] B. Jawerth and W. Sweldens, "An Overview of Wavelet Based Multiresolution Analyses," *Society for Industrial and Applied Mathematics*, vol. 36, no. 3, pp. 377- 412, 1994.
- [18] D. Flores-Tapia, Z. Moussavi and G. Thomas, "Heart Sound Cancellation Based on Multiscale Products and Linear Prediction," *IEEE Transaction On Biomedical Engineering*, vol. 54, no. 2, pp. 234 - 243, 2007.
- [19] A. Alaql, Analysis and Processing of Human Electroretinogram, *M.S.Thesis*, University of South Florida, Tampa, FL, USA 2016.